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(54)
CONTROLLER CAPABLE OF ADJUSTING SYSTEM TEMPERATURE, STORAGE DEVICE INCLUDING THE CONTROLLER, AND METHOD OF OPERATING THE STORAGE DEVICE

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See application file for complete search history.

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ABSTRACT

A storage device may migrate data stored in a partial data area of a volatile memory to a nonvolatile memory depending on a system temperature, and may deactivate the corresponding data area, whereby it is possible to variably adjust data areas on which a refresh operation is performed in the volatile memory. Since data areas on which a refresh operation is performed are adjusted depending on a system temperature, a storage device capable of adjusting the system temperature and preventing performance and reliability from degrading according to the system temperature may be provided.

18 Claims, 10 Drawing Sheets

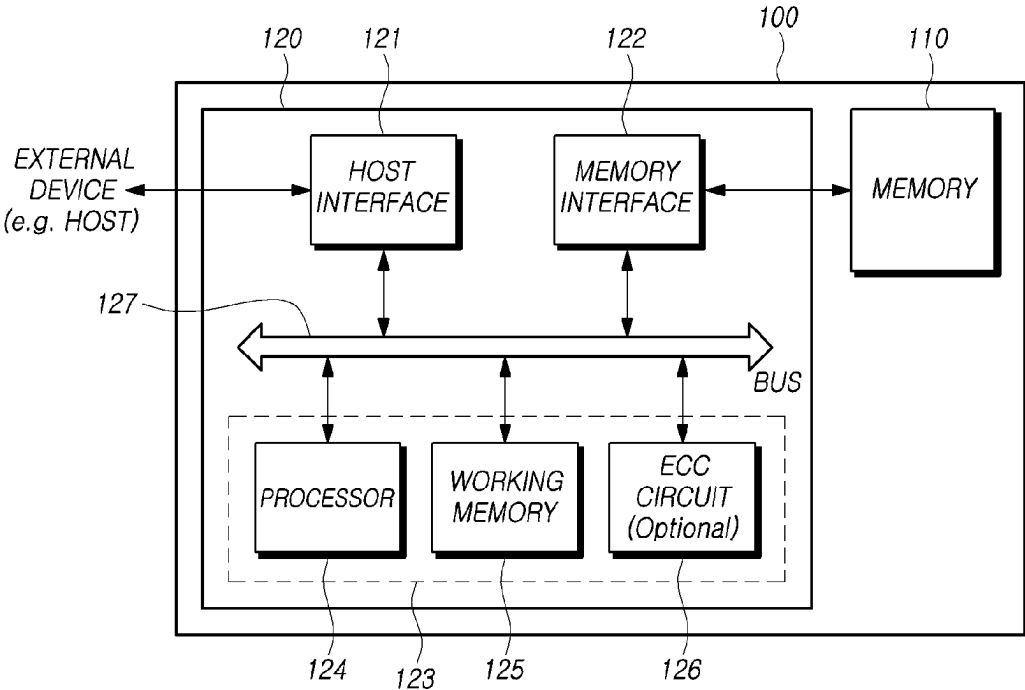


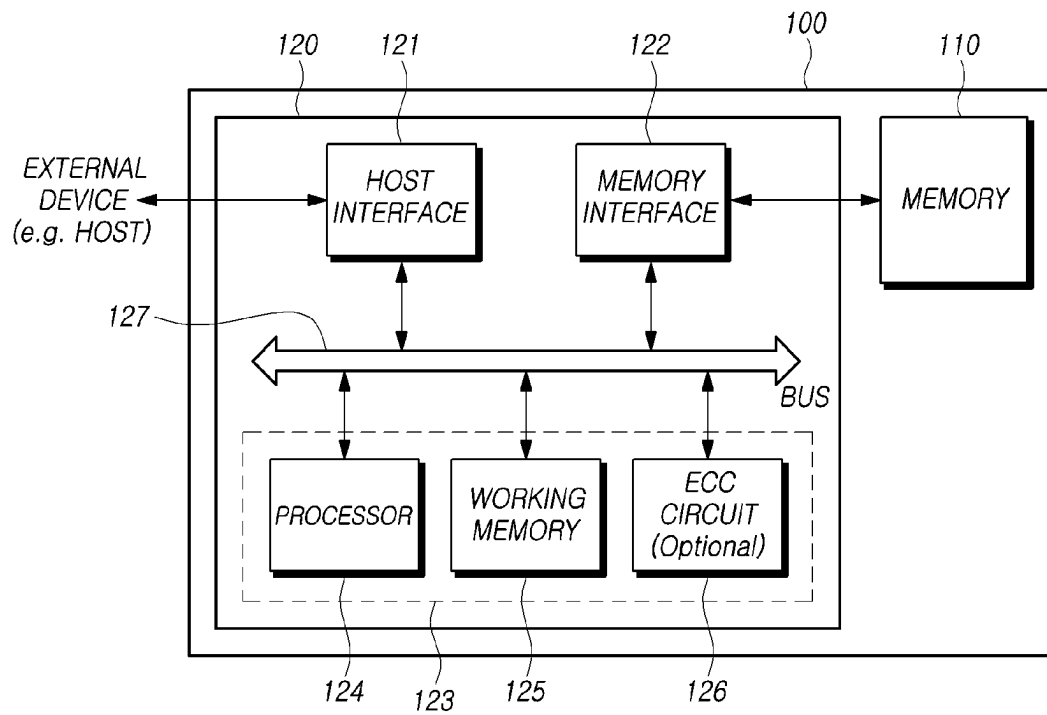
FIG. 1

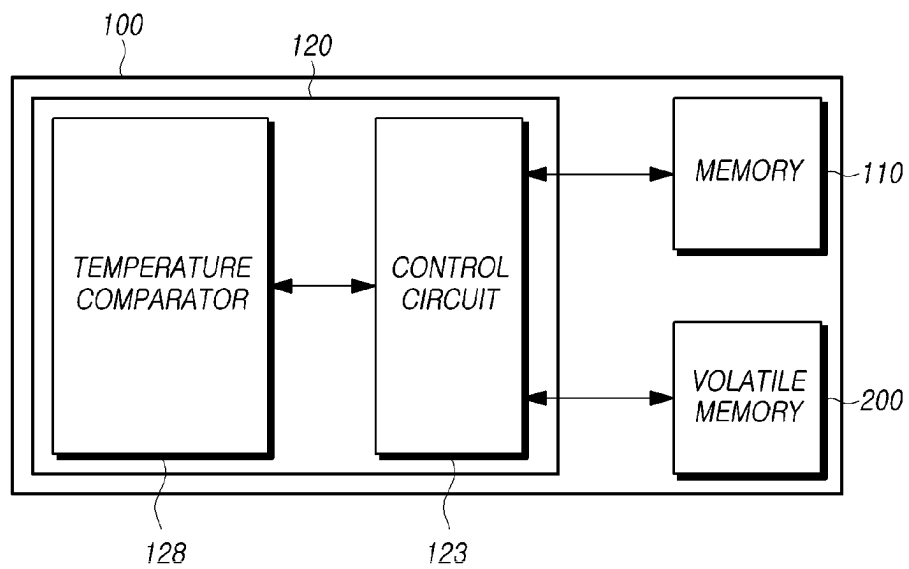
FIG. 2

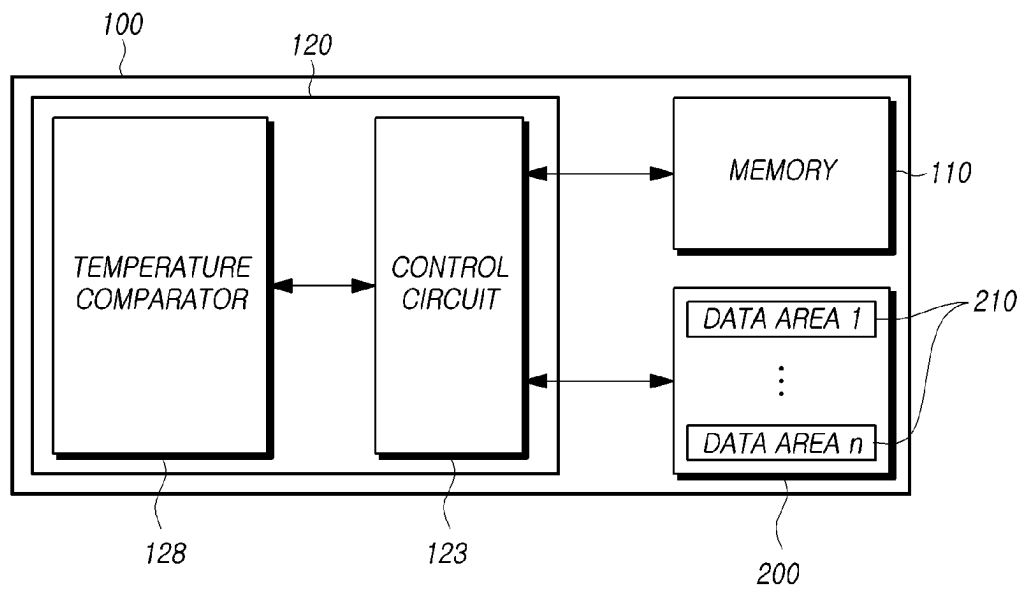
FIG. 3

FIG. 4

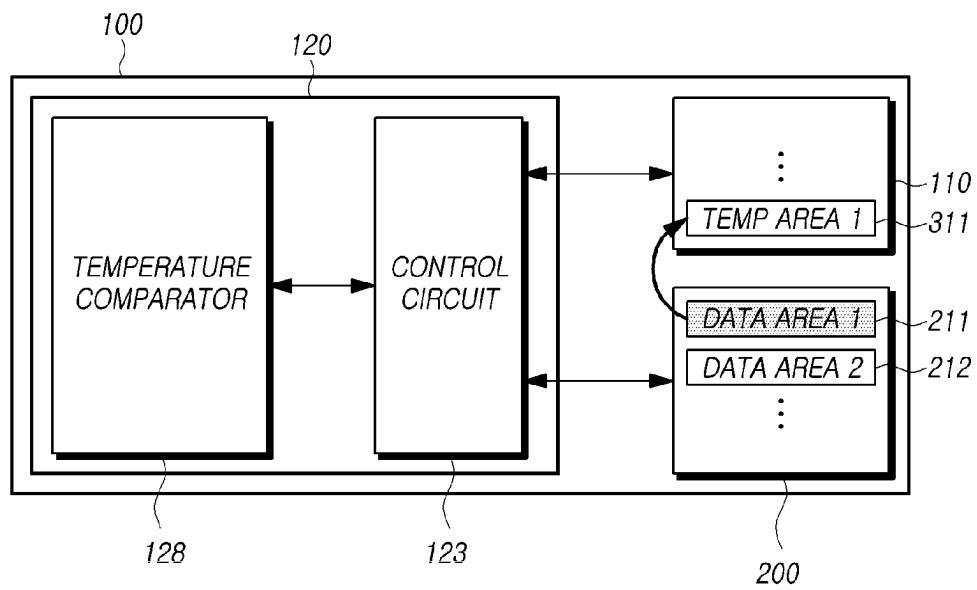


FIG. 5

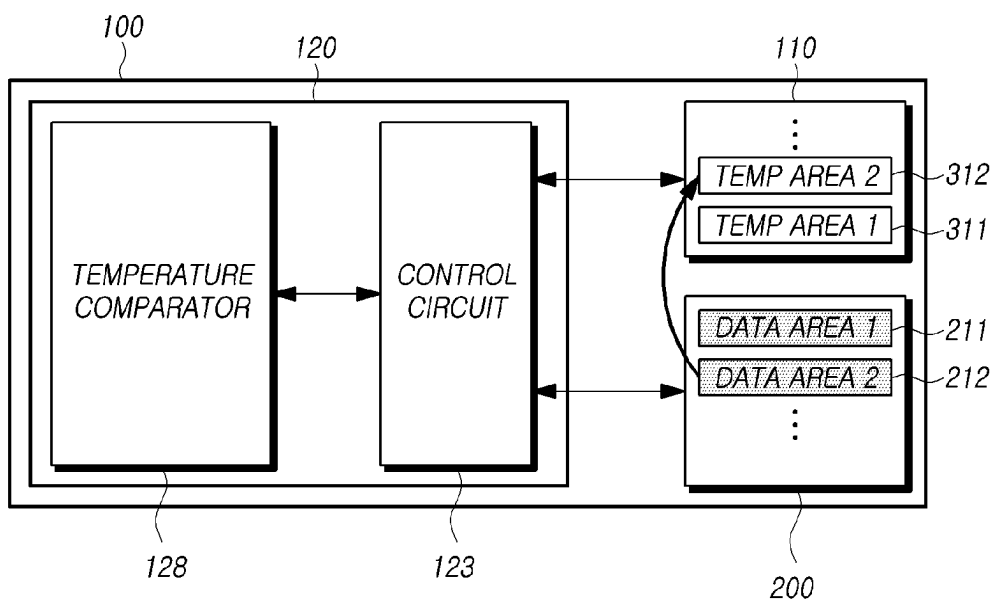


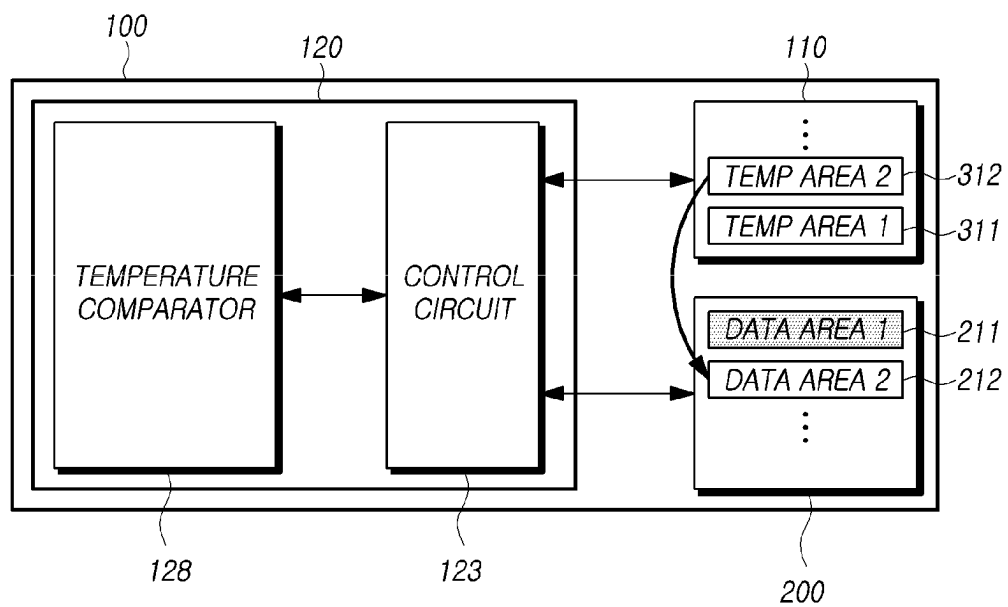
FIG. 6

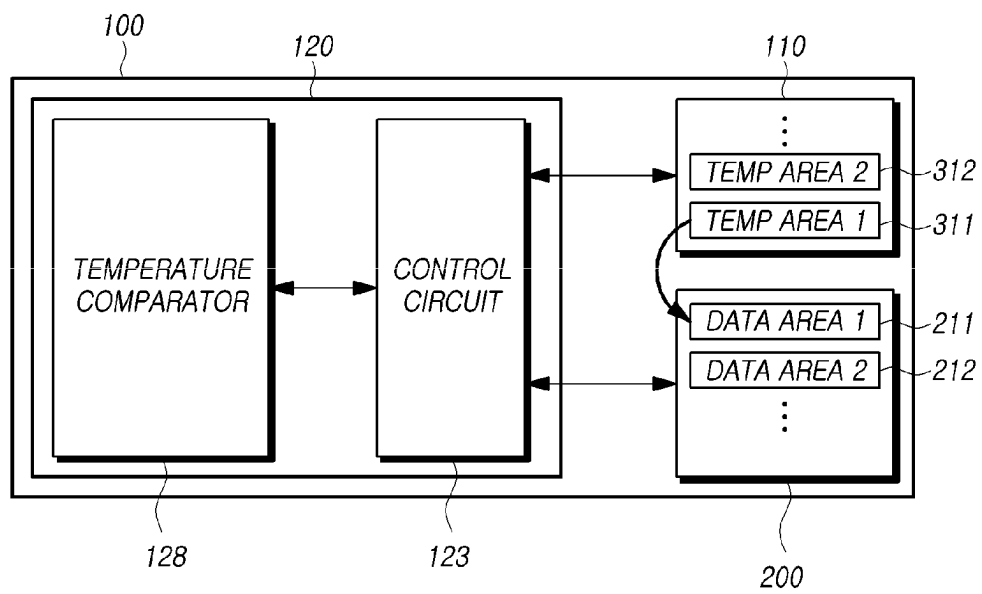
FIG. 7

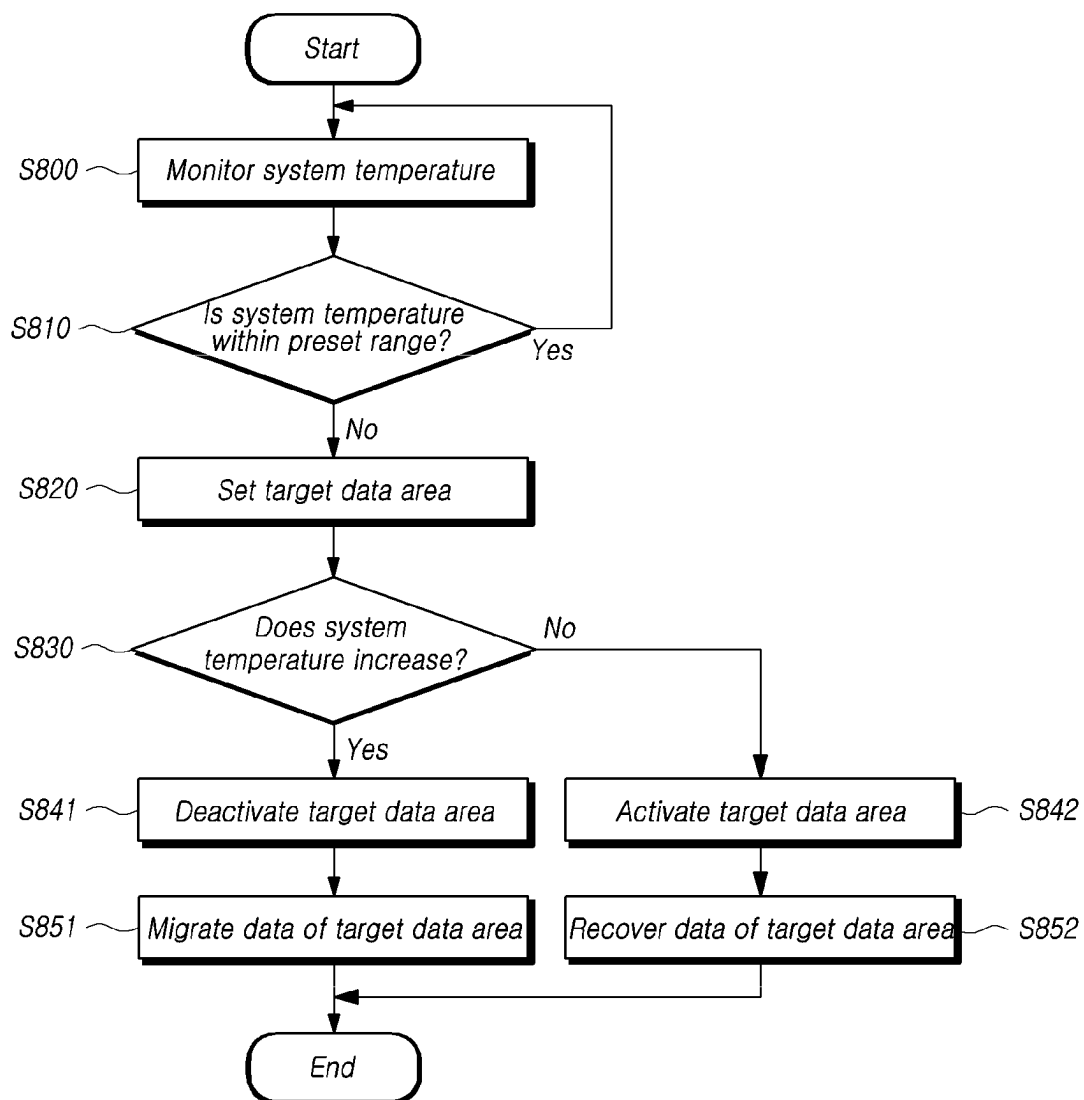
FIG. 8

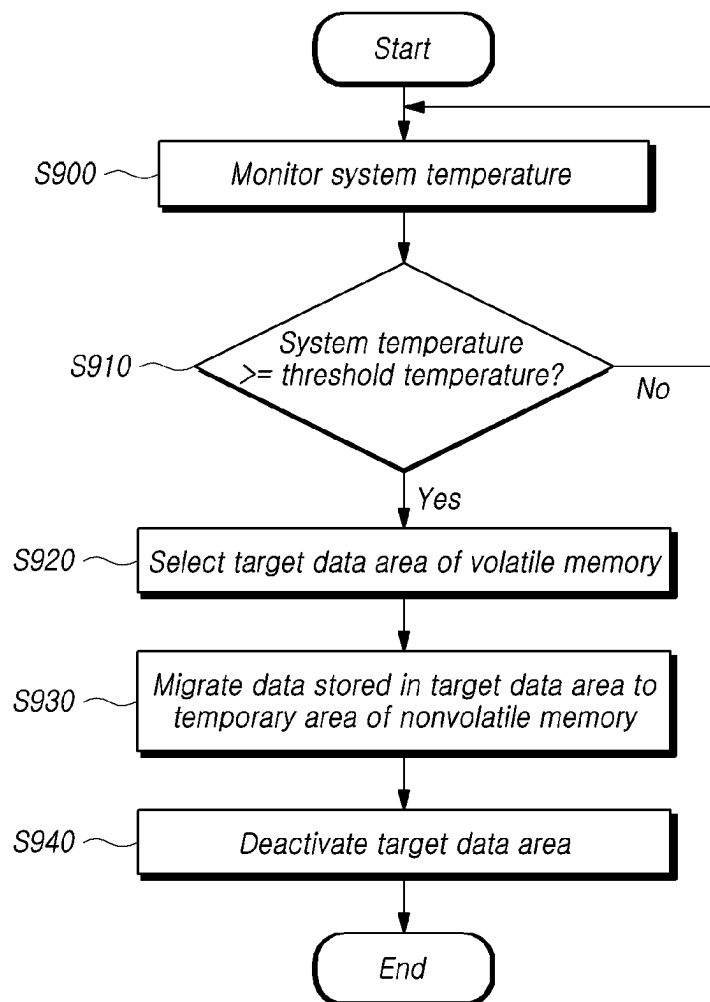
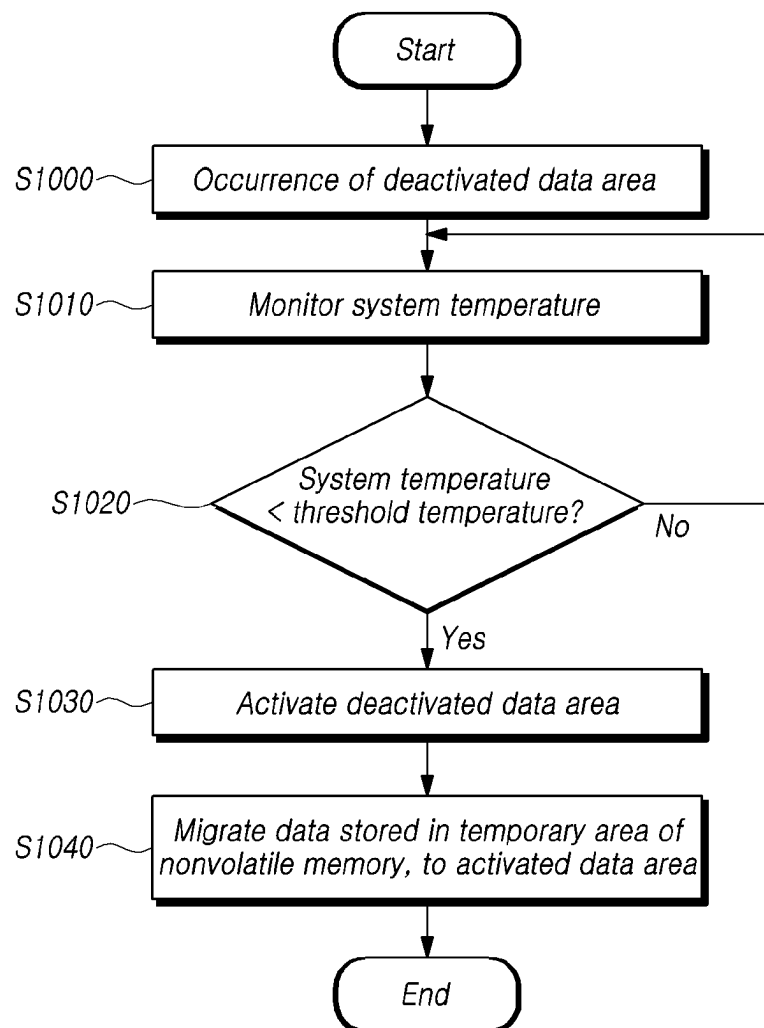
FIG. 9

FIG. 10

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CONTROLLER CAPABLE OF ADJUSTING SYSTEM TEMPERATURE, STORAGE DEVICE INCLUDING THE CONTROLLER, AND METHOD OF OPERATING THE STORAGE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

The present application claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2022-0148464 filed in the Korean Intellectual Property Office on Nov. 9, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

Various embodiments generally relate to a controller capable of adjusting a system temperature, a storage device including the controller, and a method of operating the storage device.

2. Related Art

A storage device is a device which stores data in response to a request of a host such as a computer, a mobile terminal such as a smartphone or a tablet, or any of various electronic devices.

The storage device may include a controller for controlling a memory (e.g., a volatile memory/a nonvolatile memory). The controller may receive a command from the host, and may execute or control an operation for reading, writing, or erasing data with respect to the memory included in the storage device in response to the received command.

As the storage device operates, a temperature of the storage device may increase. As the temperature of the storage device increases, performance of the storage device may degrade, or reliability of data stored in the storage device may degrade.

SUMMARY

Various embodiments are directed to providing an apparatus and a method capable of preventing degradation in performance or reliability of a storage device due to an increase in a temperature of the storage device.

In an embodiment, a storage device may include: a nonvolatile memory; a volatile memory including a plurality of data areas on at least a part of which a refresh operation is periodically performed; and a controller configured to, when a system temperature changes from a first temperature to a second temperature higher than the first temperature, store data stored in a first data area of the volatile memory in a first temporary area of the nonvolatile memory and deactivate the first data area.

In an embodiment, a storage device may include: a nonvolatile memory; a volatile memory including a plurality of data areas on at least a part of which a refresh operation is periodically performed; and a controller configured to, when a system temperature changes from a first temperature to a second temperature lower than the first temperature, activate at least a pair of data areas which are deactivated among the plurality of data areas when the system temperature is the first temperature.

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In an embodiment, a storage device may include: a nonvolatile memory; a volatile memory including a plurality of data areas on at least a part of which a refresh operation is periodically performed; and a controller configured to monitor a system temperature, set at least one of the plurality of data areas as a target data area depending on the system temperature, and adjust an activated or deactivated state of the target data area.

In an embodiment, a storage device may include: a nonvolatile memory; a volatile memory including a plurality of data areas on at least a part of which a refresh operation is periodically performed; and a controller configured to, when a system temperature is equal to or higher than a first threshold temperature, store data stored in a first data area among the plurality of data areas in a first temporary area of the nonvolatile memory and deactivate the first data area.

In an embodiment, a method for operating a storage device may include: storing, when a system temperature is equal to or higher than a threshold temperature, data stored in a target data area among a plurality of data areas of a volatile memory in a temporary area of a nonvolatile memory; and deactivating the target data area.

In an embodiment, a controller may include: a volatile memory configured to compare a system temperature with at least one preset threshold temperature; and a processor configured to, when the system temperature is equal to or higher than the at least one threshold temperature, store data stored in a partial data area of the volatile memory in an outside and deactivate the partial data area of the volatile memory.

According to the embodiments of the disclosed technology, by adjusting the temperature of the storage device even when the temperature of the storage device increases, it is possible to maintain performance and reliability of the storage device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a storage device according to an embodiment of the disclosed technology.

FIG. 2 illustrates a storage device according to another embodiment of the disclosed technology.

FIGS. 3 to 7 illustrate an operating scheme of the storage device illustrated in FIG. 2.

FIGS. 8 to 10 are flowcharts illustrating a method of operating the storage device illustrated in FIG. 2 according to an embodiment of the disclosed technology.

DETAILED DESCRIPTION

Hereinafter, embodiments of the disclosed technology will be described in detail with reference to the accompanying drawings.

FIG. 1 illustrates a storage device 100 according to an embodiment of the disclosed technology.

Referring to FIG. 1, the storage device 100 may include a memory 110 for storing data, and a controller 120 for controlling the memory 110.

The memory 110 may include a plurality of memory blocks, and operate under the control of the controller 120. Operations of the memory 110 may include a read operation, a program operation (also referred to as a write operation) and an erase operation.

The memory 110 may include a memory cell array including a plurality of memory cells (also simply referred to as “cells”) which store data. Such a memory cell array may exist in a memory block.

For example, the memory **110** may be implemented into various types such as a NAND flash memory, a 3D NAND flash memory, a NOR flash memory, a resistive random access memory (RRAM), a phase-change random access memory (PRAM), a magneto-resistive random access memory (MRAM), a ferroelectric random access memory (FRAM), and a spin transfer torque random access memory (STT-RAM).

The memory **110** may be implemented into a three-dimensional array structure. The embodiment of the disclosed technology may be applied to not only a flash memory in which a charge storage layer is configured by a conductive floating gate but also a charge trap flash (CTF) in which a charge storage layer is configured by a dielectric layer.

The memory **110** may receive a command and an address from the controller **120**, and may access an area which is selected by the address in the memory cell array. In other words, the memory **110** may perform an operation, indicated by the command, on the area selected by the address.

For example, the memory **110** may perform a program operation, a read operation, and an erase operation. In this connection, when performing the program operation, the memory **110** may program data to the area selected by the address. When performing the read operation, the memory **110** may read data from the area selected by the address. In the erase operation, the memory **110** may erase data stored in the area selected by the address.

The controller **120** may control write (or program), read, erase, and background operations for the memory **110**. For example, the background operation may include at least one among a garbage collection (GC) operation, a wear leveling (WL) operation, a read reclaim (RR) operation, a bad block management (BBM) operation, and so forth.

The controller **120** may control the operation of the memory **110** according to a request from a device (e.g., a host) located outside the storage device **100**. Also, the controller **120** may control the operation of the memory **110** regardless of a request of the host.

The host may be any of a computer, an ultra-mobile PC (UMPC), a workstation, a personal digital assistant (PDA), a tablet, a mobile phone, a smartphone, an e-book, a portable multimedia player (PMP), a portable game player, a navigation device, a black box, a digital camera, a digital multimedia broadcasting (DMB) player, a smart television, a digital audio recorder, a digital audio player, a digital picture recorder, a digital picture player, a digital video recorder, a digital video player, a storage configuring a data center, one of various electronic devices configuring a home network, one of various electronic devices configuring a computer network, one of various electronic devices configuring a telematics network, an RFID (radio frequency identification) device, a mobility device (e.g., a vehicle, a robot or a drone) capable of driving under human control or autonomous driving, etc.

The host may include at least one operating system (OS). The operating system may generally manage and control the function and operation of the host, and may provide interoperability between the host and the storage device **100**. The operating system may be classified into a general operating system and a mobile operating system depending on the mobility of the host.

The controller **120** and the host may be devices which are separated from each other. As the case may be, the controller **120** and the host may be integrated into one device. Hereunder, for the sake of convenience in explanation, it will be

described as an example that the controller **120** and the host are devices which are separated from each other.

Referring to FIG. **1**, the controller **120** may include a memory interface **122** and a control circuit **123**, and may further include a host interface **121**.

The host interface **121** provides an interface for communication with the host. For example, the host interface **121** provides an interface which uses at least one among various interface protocols such as a USB (Universal Serial Bus) protocol, an MMC (multimedia card) protocol, a PCI (Peripheral Component Interconnection) protocol, a PCI-E (PCI-express) protocol, an ATA (Advanced Technology Attachment) protocol, a serial-ATA protocol, a parallel-ATA protocol, an SCSI (Small Computer System Interface) protocol, an ESDI (Enhanced Small Disk Interface) protocol, an IDE (Integrated Drive Electronics) protocol, an SMBus (System Management Bus) protocol, an I2C (Inter-Integrated Circuit) protocol, an I3C (Improved Inter-Integrated Circuit) protocol, and a private protocol.

When receiving a command from the host, the control circuit **123** may receive the command through the host interface **121**, and may perform an operation of processing the received command.

The memory interface **122** may be coupled with the memory **110** to provide an interface for communication with the memory **110**. That is to say, the memory interface **122** may be configured to provide an interface between the memory **110** and the controller **120** under the control of the control circuit **123**.

The control circuit **123** performs the general control operation of the controller **120** to control the operation of the memory **110**. To this end, for instance, the control circuit **123** may include at least one of a processor **124** and a working memory **125**, and may selectively include an error detection and correction circuit (ECC circuit) **126**.

The processor **124** may control the general operation of the controller **120**, and may perform a logic calculation. The processor **124** may communicate with the host through the host interface **121**, and may communicate with the memory **110** through the memory interface **122**.

The processor **124** may perform the function of a flash translation layer (FTL). The processor **124** may translate a logical block address (LBA), provided by the host, into a physical block address (PBA) through the flash translation layer (FTL). The flash translation layer (FTL) may receive the logical block address (LBA) and translate it into the physical block address (PBA), by using a mapping table.

There are various address mapping methods of the flash translation layer, depending on a mapping unit. Representative address mapping methods include a page mapping method, a block mapping method, and a hybrid mapping method.

The processor **124** may randomize data received from the host. For example, the processor **124** may randomize data received from the host, by using a set randomizing seed. The randomized data may be provided to the memory **110**, and may be programmed to a memory cell array of the memory **110**.

In a read operation, the processor **124** may derandomize data received from the memory **110**. For example, the processor **124** may derandomize data received from the memory **110**, by using a derandomizing seed. The derandomized data may be outputted to the host.

The processor **124** may execute firmware to control the operation of the controller **120**. Namely, in order to control the general operation of the controller **120** and perform a logic calculation, the processor **124** may execute (drive)

firmware loaded in the working memory **125** upon booting. Hereafter, an operation of the storage device **100** to be described based on the embodiment of the disclosed technology may be implemented in such a way that the processor **124** executes firmware in which the corresponding operation is defined.

Firmware, as a program to be executed in the storage device **100** to drive the storage device **100**, may include various functional layers. For example, the firmware may include binary data in which is codes for executing the functional layers, respectively, are defined.

For example, the firmware may include at least one among a flash translation layer (FTL) which performs a translating function between a logical address requested to the storage device **100** from the host and a physical address of the memory **110**, a host interface layer (HIL) which serves to analyze a command requested to the storage device **100** from the host and transfer the command to the flash translation layer (FTL), and a flash interface layer (FIL) which transfers a command, instructed from the flash translation layer (FTL), to the memory **110**.

Such firmware may be loaded in the working memory **125** from, for example, the memory **110** or a separate nonvolatile memory (e.g., a ROM or a NOR Flash) located outside the memory **110**. The processor **124** may first load all or a part of the firmware in the working memory **125** when executing a booting operation after power-on.

The processor **124** may perform a logic calculation which is defined in the firmware loaded in the working memory **125**, to control the general operation of the controller **120**. The processor **124** may store a result of performing the logic calculation defined in the firmware, in the working memory **125**. The processor **124** may control the controller **120** to generate a command or a signal, according to a result of performing the logic calculation defined in the firmware. When a part of firmware in which a logic calculation to be performed is defined is not loaded in the working memory **125**, the processor **124** may generate an event (e.g., an interrupt) for loading the corresponding part of the firmware in the working memory **125**.

The processor **124** may load metadata necessary for driving firmware, from the memory **110**. The metadata, as data for managing the memory **110**, may include management information on user data stored in the memory **110**.

Firmware may be updated while the storage device **100** is manufactured or while the storage device **100** is executed. The controller **120** may download new firmware from the outside of the storage device **100** and update existing firmware with the new firmware.

The working memory **125** may store firmware, a program code, a command and data which are necessary to drive the controller **120**. Such a working memory **125** as, for example, a volatile memory, may include at least one among an SRAM (static RAM), a DRAM (dynamic RAM) and an SDRAM (synchronous DRAM).

The error detection and correction circuit **126** may detect an error bit of target data and correct the detected error bit, by using an error correction code. The target data may be, for example, data stored in the working memory **125** or data read from the memory **110**.

The error detection and correction circuit **126** may be implemented to decode data by using the error correction code. The error detection and correction circuit **126** may be implemented by various code decoders. For example, a decoder which performs unsystematic code decoding or a decoder which performs systematic code decoding may be used.

For example, the error detection and correction circuit **126** may detect an error bit by the unit of a set sector in each of read data. Namely, each read data may be constituted by a plurality of sectors. A sector may mean a data unit smaller than a page being the read unit of a flash memory. Sectors constituting each read data may be matched with one another by the medium of an address.

The error detection and correction circuit **126** may calculate a bit error rate (BER), and may determine whether an error is correctable or not, by the unit of a sector. For example, when a bit error rate (BER) is higher than a reference value, the error detection and correction circuit **126** may determine that a corresponding sector is uncorrectable or a fail. On the other hand, when a bit error rate (BER) is lower than the reference value, the error detection and correction circuit **126** may determine that a corresponding sector is correctable or a pass.

The error detection and correction circuit **126** may perform an error detection and correction operation sequentially for all read data. When a sector included in read data is correctable, the error detection and correction circuit **126** may omit an error detection and correction operation for a corresponding sector for next read data. When the error detection and correction operation for all read data is ended in this way, the error detection and correction circuit **126** may detect a sector which is determined to be uncorrectable to the last. There may be one or more sectors that are determined to be uncorrectable. The error detection and correction circuit **126** may transfer information (e.g., address information) on a sector which is determined to be uncorrectable, to the processor **124**.

A bus **127** may be configured to provide channels among the components **121**, **122**, **124**, **125**, and **126** of the controller **120**. The bus **127** may include, for example, a control bus for transferring various control signals, commands and the likes, a data bus for transferring various data, and so forth.

Some components among the above-described components **121**, **122**, **124**, **125**, and **126** of the controller **120** may be omitted, or some components among the above-described components **121**, **122**, **124**, **125**, and **126** of the controller **120** may be integrated into one component. As the case may be, in addition to the above-described components **121**, **122**, **124**, **125**, and **126** of the controller **120**, one or more other components may be added.

During an operation of the memory **110** and the storage device **100** including the memory **110**, a temperature of the storage device **100** may increase. When the temperature of the storage device **100** increases, performance and reliability of the storage device **100** may degrade.

Embodiments of the disclosed technology may provide an apparatus and a method capable of maintaining the performance and the reliability of the storage device **100** by adjusting the temperature of the storage device **100** when the temperature of the storage device **100** increases.

FIG. 2 illustrates a storage device **100** according to another embodiment of the disclosed technology. FIGS. 3 to 7 illustrate an operating scheme of the storage device **100** illustrated in FIG. 2 according to an embodiment of the disclosed technology.

Referring to FIG. 2, the storage device **100** may include a memory **110**, a controller **120**, and a volatile memory **300**. The memory **110** may be a nonvolatile memory.

The controller **120** may include at least some of the components described above with reference to FIG. 1. The controller **120** may include a control circuit **123** and a temperature comparator **128**.

The volatile memory **200** may be one of memories which are located inside the storage device **100**.

The volatile memory **200** may be located inside the controller **120** or outside the controller **120**.

When the volatile memory **200** is located inside the controller **120**, the volatile memory **200** may be the working memory **125** illustrated in FIG. 1. Alternatively, the volatile memory **200** may be another memory which is included in the controller **120** in addition to the working memory **125**.

The temperature comparator **128** may compare a system temperature with at least one preset threshold temperature. The system temperature may be a temperature of the storage device **100**. The system temperature may be a temperature of at least one component included in the storage device **100**. Therefore, the system temperature may be a temperature of at least one of the memory **110**, the controller **120**, and the volatile memory **200**.

Based on a comparison result of the temperature comparator **128**, the control circuit **123** may perform an operation for adjusting the system temperature. In an embodiment, the operation for adjusting the system temperature may be performed by a processor included in the control circuit **123**, e.g., the processor **124** illustrated in FIG. 1.

For example, when the system temperature is equal to or higher than the threshold temperature, the controller **120** may delay and/or stop a certain operation of the storage device **100**.

For example, the controller **120** may delay a background operation. Alternatively, the controller **120** may delay processing of a command inputted from a host.

When the system temperature is equal to or higher than the threshold temperature, the controller **120** may stop some control operations to be performed on the volatile memory **200**.

For example, when the system temperature is equal to or higher than the threshold temperature, the controller **120** may designate a certain data area among a plurality of data areas included in the volatile memory **200** as a target data area.

The controller **120** may migrate data stored in the target data area of the volatile memory **200** to the memory **110**.

When the data stored in the target data area of the volatile memory **200** is stored in the memory **110**, the controller **120** may deactivate the target data area of the volatile memory **200**.

When the target data area of the volatile memory **200** is deactivated, a refresh operation on the target data area may be stopped. That is, the refresh operation may not be performed on the target data area.

The refresh operation may be periodically performed to prevent data stored in memory cells of the volatile memory **200** from being damaged by a leakage current. Since the data stored in the target data area is migrated to and stored in the memory **110**, the refresh operation on the target data area may be stopped. However, the refresh operation may be periodically performed on data areas other than the target data area.

When the system temperature is equal to or higher than the threshold temperature, by stopping performing the refresh operation on the target data area, it is possible to reduce the system temperature or reduce a degree of increase in the system temperature.

For example, when the system temperature increases from a first temperature to a second temperature higher than the first temperature, the size (or number) of data areas on which the refresh operation is performed in the volatile memory **200** may decrease. The size of data areas on which the

refresh operation is performed at the second temperature may be smaller than the size of data areas on which the refresh operation is performed at the first temperature.

When the system temperature increases from the second temperature to a third temperature higher than the second temperature, the size of data areas on which the refresh operation is performed in the volatile memory **200** may further decrease. That is, as the system temperature increases, the size of data areas on which the refresh operation is performed may gradually decrease.

Since the data stored in the target data area is migrated to and stored in the memory **110**, the controller **120** should read the data from the memory **110** when the data is required to perform a necessary operation relating to the data.

In this way, when the system temperature increases, the controller **120** may adjust the system temperature by delaying a certain operation and/or stopping performing a refresh operation on some data areas of the volatile memory **200**.

The operation of delaying the certain operation and/or the operation of stopping performing the refresh operation on some data areas may be performed according to the same standard or may be performed according to different standards.

In an embodiment, when the system temperature is equal to or higher than the threshold temperature, the controller **120** may delay the certain operation of the storage device **100** and simultaneously stop performing the refresh operation on some data areas of the volatile memory **200**.

In another embodiment, when the system temperature is equal to or higher than a reference temperature lower than the threshold temperature, but is lower than the threshold temperature, the controller **120** may delay the certain operation of the storage device **100**. When the system temperature is equal to or higher than the threshold temperature, the controller **120** may stop performing the refresh operation on some data areas.

In still another embodiment, in order to adjust the system temperature, the controller **120** may delay the certain operation of the storage device **100** or stop performing the refresh operation on some data areas.

The controller **120** may select a certain data area, as the target data area on which the refresh operation is not performed, in consideration of data stored in the certain data area.

In an embodiment, a data area storing data, which has a relatively small influence on a system operation even when being migrated to and stored in the memory **110**, among data stored in the volatile memory **200**, may be selected as the target data area.

In another embodiment, a data area, which has a longest elapse time after data stored in the data area has been updated, among the plurality of data areas included in the volatile memory **200** may be selected as the target data area.

In still another embodiment, a data area, which has the smallest number of data updating times for a preset time period (referred to as a "data updating count value"), among the plurality of data areas included in the volatile memory **200**, may be selected as the target data area.

In further still another embodiment, a data area, which stores data having a low use frequency or being temporarily not used, may be selected as the target data area.

Besides, the target data area may be selected in various ways in consideration of characteristics of stored data.

In an embodiment, the target data area may be one of a set of data areas in the volatile memory **200**, and the set of data areas are physically distinguished from each other.

In another embodiment, the target data area may be one of a set of data areas in the volatile memory **200**, and the set of data areas are distinguished from each other according to attributes (e.g., Map, Log, etc.) of data stored therein. According to an attribute of data, a data area storing data that has an attribute of a low use frequency may be selected as the target data area.

When the system temperature becomes lower than the threshold temperature after the target data area has been deactivated, the controller **120** may activate the deactivated target data area. If necessary, the controller **120** may migrate the data, which has been migrated to and stored in the memory **110**, to the target data area again.

For example, when the system temperature decreases from a first temperature to a second temperature lower than the first temperature, at least some of the data areas of the volatile memory **200** which have been deactivated when the system temperature was the first temperature may be activated. A refresh operation may be periodically performed on a data area activated at the second temperature. The size of data areas on which the refresh operation is periodically performed in the volatile memory **200** when the system temperature is the first temperature may be equal to or smaller than the size of data areas on which the refresh operation is periodically performed in the volatile memory **200** when the system temperature is the second temperature.

When the system temperature decreases from the second temperature to a third temperature lower than the second temperature, the size of data areas on which the refresh operation is performed in the volatile memory **200** may further increase. That is, as the system temperature decreases, the size of data areas on which the refresh operation is performed may gradually increase. The size of data areas which are deactivated and thus on which the refresh operation is not performed may decrease as the system temperature decreases.

Since the controller **120** selectively performs the refresh operation on some data areas of the volatile memory **200** according to the system temperature, the system temperature of the storage device **100** may be easily adjusted, and thus it is possible to prevent the performance or reliability of the storage device **100** from degrading due to an increase in the system temperature of the storage device **100**.

The aforementioned control operation may be performed by the control circuit **123** included in the controller **120**, e.g., by the processor **124** in the control circuit **123**, but embodiments are not limited thereto.

The controller **120** may control the refresh operation based on one threshold temperature, or may control the refresh operation based on at least two threshold temperatures.

FIGS. 3 to 7 illustrate an operating scheme of the storage device **100** illustrated in FIG. 2.

Referring to FIG. 3, the storage device **100** may include the volatile memory **200**. The volatile memory **200** may include a plurality of data areas **210**. The plurality of data areas **210** may be physically distinguished from each other. Alternatively, the plurality of data areas **210** may be distinguished from each other according to attributes of data to be stored therein.

The storage device **100** may include the memory **110** and the controller **120**. The controller **120** may include the temperature comparator **128** and the control circuit **123**.

The temperature comparator **128** may compare the system temperature with at least one preset threshold temperature.

The temperature comparator **128** may measure, as the system temperature, a temperature of at least one of the

components included in the storage device **100**, e.g., a temperature of at least one of the memory **110**, the controller **120**, and the volatile memory **200**, using a sensor. Alternatively, the temperature comparator **128** may receive the system temperature from the outside.

The control circuit **123** may adjust the system temperature based on a temperature comparison result of the temperature comparator **128**.

For example, referring to FIG. 4, when the system temperature is equal to or higher than a first threshold temperature, the control circuit **123** may select a target data area among the plurality of data areas **210** included in the volatile memory **200**. For example, a data area which stores data having the lowest use frequency may be selected as the target data area.

FIG. 4 illustrates a case where a first data area **211** among the plurality of data areas **210** included in the volatile memory **200** is selected as the target data area.

The control circuit **123** may migrate data stored in the first data area **211** to a first temporary area **311** of the memory **110**. The first temporary area **311** may be a free block in the memory **110**, the free block storing no data. Alternatively, the first temporary area **311** may be a part of an area in which data is not stored, among areas of the memory **110** that are allocated to store information necessary for controlling the storage device **100**.

The control circuit **123** may deactivate the first data area **211** after migrating the data stored in the first data area **211** to the first temporary area **311**. The control circuit **123** may manage information on the physical location of the first data area **211** in the volatile memory **200**. For example, the control circuit **123** may manage mapping information between the physical location of the first data area **211** and the physical location of the first temporary area **311**. The aforementioned information, e.g., the mapping information, may be stored in a working memory included in the control circuit **123**, e.g., the working memory **125** shown in FIG. 1. The aforementioned information may be retained during a period in which the first data area **211** is deactivated.

The control circuit **123** may stop performing a refresh operation on the first data area **211** when the first data area **211** is deactivated. As the case may be, the control circuit **123** may delete the data stored in the first data area **211**.

Although the refresh operation performed on the first data area **211** is stopped, a refresh operation on a second data area **212** that is still activated may be periodically performed.

Since the total number of refresh operations performed in the volatile memory **200** decreases, the system temperature may decrease or a degree of increase in the system temperature may decrease.

Thereafter, the temperature comparator **128** may continuously or periodically perform a comparison between the system temperature and the first threshold temperature.

The temperature comparator **128** may perform a comparison between the system temperature and a second threshold temperature that is higher than the first threshold temperature.

Referring to FIG. 5, after the first data area **211** is deactivated, when the system temperature is equal to or higher than the second threshold temperature, the control circuit **123** may further select the second data area **212** among the plurality of data areas **210** included in the volatile memory **200**, as a target data area. The second data area **212** may be in the same memory block as the first data area **211** or in a different memory block from the first data area **211**. The second data area **212** may be selected as the target data

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area according to the same standard as the standard for selecting the first data area **211** as the target data area.

The control circuit **123** may migrate data stored in the second data area **212** to a second temporary area **312** of the memory **110**. The second temporary area **312** may be a free block in the memory **110**. The second temporary area **212** may be in the same memory block as the first temporary area **211** or in a different memory block from the first temporary area **211**.

The control circuit **123** may deactivate the second data area **212** after the data stored in the second data area **212** is migrated to the second temporary area **312**. A refresh operation performed on the second data area **212** may be stopped when the second data area **212** is deactivated. As the case may be, the data stored in the second data area **212** may be deleted. The control circuit **123** may manage information on the physical location of the second data area **212**, e.g., mapping information between the physical location of the second data area **212** and the physical location of the second temporary area **312** while the second data area **212** is deactivated.

As the system temperature becomes equal to or higher than the second threshold temperature that is higher than the first threshold temperature, the number of data areas on which a refresh operation is not performed in the volatile memory **200** may be increased.

The second data area **212** may be selected according to the use frequency of data stored therein.

For example, the second data area **212** may be a data area whose elapse time after the latest updating of the stored data is shorter than an elapse time after the latest updating of the first data area **211**. The second data area **212** may have the elapse time after the latest updating that is longer than elapse times of the remaining data areas **210** other than the first data area **211**.

Alternatively, the second data area **212** may be a data area whose data updating count value for a preset time period is greater than a data updating count value of the first data area **211** for the preset time period. The data updating count value of the second data area **212** may be smaller than data updating count values of the remaining data areas **210** for the preset time period.

The second data area **212** may store data having a use frequency that is higher than a use frequency of the data stored in the first data area **211** and lower than use frequencies of data stored in the remaining data areas **210**.

The size of the second data area **212** may be the same as that of the first data area **211**. Alternatively, the size of the second data area **212** may be larger than that of the first data area **211**. As the system temperature increases, a degree of increase in the number of deactivated data areas may increase.

As such, the control circuit **123** may control the size of data areas to be deactivated in the volatile memory **200** based on at least two threshold temperatures. The control circuit **123** may increase the number of data areas to be deactivated by comparing the system temperature and a third threshold temperature higher than the second threshold temperature.

The control circuit **123** may delay a certain operation of the storage device **100** simultaneously with or separately from deactivating some of the data areas **210** of the volatile memory **200**.

In an embodiment, when the system temperature is lower than the first threshold temperature and is equal to or higher than a reference temperature, the control circuit **123** may

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delay at least one of a background operation and a processing of a command inputted from the host.

In another embodiment, when the system temperature is equal to or higher than the first threshold temperature, the control circuit **123** may delay at least one of a background operation and a processing of a command inputted from the host.

In still another embodiment, when the system temperature is equal to or higher than the first threshold temperature, the control circuit **123** may deactivate a certain data area in the volatile memory **200**. When the system temperature is equal to or higher than the second threshold temperature, the control circuit **123** may delay at least one of a background operation and a processing of a command inputted from the host in addition to deactivating the certain data area. That is, when the system temperature is equal to or higher than the second threshold temperature higher than the first threshold temperature, the control circuit **123** may delay the certain operation and deactivate the certain data area of the volatile memory **200**.

The delaying of the background operation and the delaying of the processing of the command inputted from the host are examples. The control circuit **123** may delay at least a certain operation among operations other than a refresh operation according to the system temperature.

After the first data area **211** and the second data area **212** of the volatile memory **200** are deactivated, the control circuit **123** may activate at least one of the deactivated first and second data areas **211** and **212** again according to a change in the system temperature.

Referring to FIG. 6, after the first data area **211** and the second data area **212** are deactivated, the system temperature may become equal to or higher than the first threshold temperature and lower than the second threshold temperature.

When the system temperature becomes lower than the second threshold temperature, the control circuit **123** may activate the second data area **212** of the volatile memory **210** so that the second data area **212** is changed to a state capable of storing data. Thus, a refresh operation may be periodically performed on the second data area **212**.

As the case may be, the control circuit **123** may migrate the data stored in the second temporary area **312** of the memory **110** to the second data area **212**.

Since the system temperature is lower than the second threshold temperature and is equal to or higher than the first threshold temperature, the control circuit **123** may activate the second data area **212**, and may maintain the first data area **211** in the deactivated state.

Thereafter, the first data area **211** may be also activated according to a change in the system temperature.

Referring to FIG. 7, when the system temperature becomes lower than the first threshold temperature, the control circuit **123** may also activate the first data area **211** of the volatile memory **200**. Thus, data can be stored in the first data area **211**, and a refresh operation may be periodically performed on the first data area **211**.

As the case may be, the control circuit **123** may migrate the data stored in the first temporary area **311** of the memory **110** to the first data area **211**.

According to the embodiment of the disclosed technology, since the size of data areas on which a refresh operation is periodically performed is adjusted depending on the system temperature, it is possible to adjust the system temperature and thus prevent the performance or the reliability of the storage device **100** from degrading due to the system temperature.

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FIGS. 8 to 10 are flowcharts illustrating a method of operating the storage device **100** of FIG. 2 according to an embodiment of the disclosed technology. The method will be described with reference to FIG. 2.

Referring to FIG. 8, the controller **120** may continuously or periodically monitor a system temperature of the storage device **100** (S800).

The controller **120** may check whether the system temperature is within a preset range (S810). The controller **120** may continuously monitor the system temperature to thereby determine whether the system temperature is within the preset range.

When the system temperature is out of the preset range, the controller **120** may set a data area included in the volatile memory **200** as a target data area (S820).

For example, when the system temperature increases beyond the preset range, the controller **120** may set a certain data area in an activated state as the target data area. For another example, when the system temperature decreases to be lower than the preset range, the controller **120** may set a certain data area in a deactivated state as the target data area.

The controller **120** may change the state of the certain data area selected as the target data area according to an increase or a decrease in the system temperature.

For example, the controller **120** may check whether the system temperature increases (S830). When the system temperature increases to be higher than a threshold temperature, the controller **120** may select the certain data area in the activated state as the target data area and change the target data area to a deactivated state (S841). When the system temperature decreases to be lower than the threshold temperature, the controller **120** may select the certain data area in the deactivated state as the target data area and change the target data area to an activated state (S842).

The controller **120** may migrate data related with the target data area whose state is changed, to the memory **110** or to the volatile memory **200**.

For example, when the target data area is deactivated, the controller **120** may migrate data stored in the target data area to the memory **110** (S851). When the target data area is activated, the controller **120** may recover data stored in the memory **110** to the target data area in the volatile memory **200** (S852). The data recovered to the target data area may be the data which is previously stored in the target data area and is migrated to and stored in the memory **110** as the target data area is changed from the activated state to the deactivated state.

The controller **120** may stop performing a refresh operation on a deactivated data area in the volatile memory **200**, and may perform the refresh operation only on an activated data area. Since the size of data areas on which the refresh operation is performed is adjusted by monitoring the system temperature, when the system temperature increases, it is possible to easily lower the system temperature and maintain the system temperature at an appropriate level.

The method of operating the storage device **100** when the system temperature increases and decreases will be described below.

Referring to FIG. 9, the controller **120** may continuously or periodically monitor the system temperature of the storage device **100** (S900).

The controller **120** may compare the system temperature with a threshold temperature (S910).

When the system temperature is equal to or higher than the threshold temperature, the controller **120** may select, as a target data area, a certain data area in the volatile memory **200** (S920). A data area which stores data having a low use

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frequency among data areas included in the volatile memory **200** may be selected as the target data area. In the volatile memory **200**, the data areas may be physically distinguished from each other, or may be distinguished from each other according to attributes of data stored therein.

The controller **120** may migrate data stored in the target data area to a temporary area of a nonvolatile memory, e.g., the memory **110** (S930).

The controller **120** may deactivate the target data area whose data is migrated to the memory **110** (S940).

When the data area selected as the target data area is deactivated, a refresh operation may not be performed on the deactivated target data area, whereas the refresh operation may be periodically performed on data areas other than the target data area in the volatile memory **200**.

After the data area selected as the target data area is deactivated, the controller **120** may increase or decrease the number of deactivated data areas in the volatile memory **200** according to a change in the system temperature.

Referring to FIG. 10, after the certain data area is deactivated (S1000), the controller **120** may continuously or periodically monitor the system temperature of the storage device **100** (S1010).

The controller **120** may compare the system temperature with the threshold temperature (S1020).

When the system temperature is lower than the threshold temperature, the controller **120** may activate the deactivated data area (S1030). Data may be stored in the activated data area, and a refresh operation may be periodically performed on the activated data area again.

As the case may be, the controller **120** may migrate the data, which has been migrated to and stored in the temporary area of the nonvolatile memory **110**, to the activated data area (S1040).

By comparing the system temperature with at least two threshold temperatures, the controller **120** may selectively adjust the size of data areas to be deactivated in the volatile memory **200**. The controller **120** may delay a certain operation other than the refresh operation depending on the system temperature simultaneously with or separately from deactivating some of data areas in the volatile memory **200**.

According to the embodiment of the disclosed technology, the size of data areas on which the refresh operation is performed may be reduced by deactivating some of the data areas of the volatile memory **200** depending on the system temperature.

Since a data area to be deactivated is selected based on the use frequency of stored data, it is possible to prevent an operational abnormality due to deactivation of the data area.

By migrating data stored in the deactivated data area to the nonvolatile memory **110**, the corresponding data can be used when it is necessary.

Because the size of data areas on which the refresh operation is performed in the volatile memory **200** is variably adjusted depending on a change in the system temperature, it is possible to control the system temperature and thus prevent the performance and the reliability of the storage device **100** from degrading due to an increase in the system temperature.

Although various embodiments of the disclosed technology have been described with particular specifics and varying details for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions may be made based on what is disclosed or illustrated in the present disclosure without departing from the spirit and scope of the invention as defined in the following claims.

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What is claimed is:

1. A storage device comprising:

a nonvolatile memory;

a volatile memory including a plurality of data areas; and
a controller configured to, when a system temperature
changes from a first temperature to a second tempera-
ture higher than the first temperature, migrate data
stored in a first data area of the volatile memory to a
first temporary area of the nonvolatile memory, which
is a different type than the volatile memory, and deac-
tivate the first data area,

wherein the controller controls a size of the first data area
to be deactivated in the volatile memory based on a
comparison between the system temperature and at
least two threshold temperature values.

2. The storage device according to claim 1, wherein, when
the first data area is deactivated, a refresh operation is not
performed on the first data area, and the refresh operation is
periodically performed on data areas other than the first data
area.

3. The storage device according to claim 1, wherein, in the
volatile memory, a size of data areas on which a refresh
operation is periodically performed when the system tem-
perature is the second temperature is smaller than a size of
data areas on which the refresh operation is periodically
performed when the system temperature is the first tempera-
ture.

4. The storage device according to claim 1, wherein, when
the system temperature changes from the second tempera-
ture to a third temperature higher than the second tempera-
ture, the controller migrates data stored in a second data area
of the volatile memory to a second temporary area of the
nonvolatile memory and deactivates the second data area in
addition to deactivating the first data area.

5. The storage device according to claim 1, wherein, when
the system temperature changes from the second tempera-
ture to the first temperature, the controller activates the first
data area that is in a deactivated state.

6. A storage device comprising:

a nonvolatile memory;

a volatile memory including a plurality of data areas; and
a controller configured to, when a system temperature
changes from a first temperature to a second tempera-
ture lower than the first temperature, activate a first data
area deactivated when the system temperature is the
first temperature,

wherein the controller controls a size of the first data area
to be deactivated in the volatile memory based on a
comparison between the system temperature and at
least two threshold temperature values.

7. The storage device according to claim 6, wherein a
refresh operation is periodically performed on the first data
area that is activated when the system temperature is
changed to the second temperature.

8. The storage device according to claim 6, wherein a
refresh operation is not performed on the first data area that
is deactivated when the system temperature is the first
temperature.

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9. The storage device according to claim 6, wherein, in the
volatile memory, a size of data areas on which a refresh
operation is periodically performed when the system tem-
perature is the first temperature is equal to or smaller than a
size of data areas on which the refresh operation is periodi-
cally performed when the system temperature is the second
temperature.

10. The storage device according to claim 6, wherein the
controller migrates data stored in a temporary area of the
nonvolatile memory to the first data area that is activated
when the system temperature is the second temperature.

11. The storage device according to claim 6, wherein,
when the system temperature changes from the second
temperature to a third temperature lower than the second
temperature, the controller activates at least one of data areas
deactivated when the system temperature is the second
temperature.

12. A storage device comprising:

a nonvolatile memory;

a volatile memory including a plurality of data areas; and
a controller configured to monitor a system temperature,
and set at least one of the plurality of data areas as a
target data area and change an activated or deactivated
state of the target data area, according to the system
temperature,

wherein the controller controls a size of the target data
area to be deactivated in the volatile memory based on
a comparison between the system temperature and at
least two threshold temperature values.

13. The storage device according to claim 12, wherein,
when the system temperature increases to be equal to or
higher than a first temperature, the controller sets, as the
target data area, at least one of data areas which are in an
activated state among the plurality of data areas, and deac-
tivates the target data area.

14. The storage device according to claim 13, wherein,
when the target data area is deactivated, the controller stops
performing a refresh operation on the target data area.

15. The storage device according to claim 13, wherein,
when the target data area is deactivated, the controller
migrates data stored in the target data area to the nonvolatile
memory.

16. The storage device according to claim 12, wherein,
when the system temperature decreases to be lower than a
second temperature, the controller sets, as the target data
area, at least one of data areas which are in a deactivated
state among the plurality of data areas, and activates the
target data area.

17. The storage device according to claim 16, wherein,
when the target data area is activated, the controller peri-
odically performs a refresh operation on the target data area.

18. The storage device according to claim 12, wherein a
size of data areas on which a refresh operation is periodically
performed among the plurality of data areas is inversely
proportional to the system temperature.

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